

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
 B &:= A^* \cup (A^* + c) \\
 C &:= (A^* + 7c - d) \cup (A^* + 8c - d) \\
 D &:= (A^* + 10c - d) \cup (A^* + 11c - d) \\
 E &:= (A^* + 17c - 2d) \cup (A^* + 18c - 2d) \\
 F &:= (A^* + 30c - 3d) \cup (A^* + 31c - 3d) \\
 G &:= (A^* + 37c - 4d) \cup (A^* + 38c - 4d) \\
 H &:= (A^* + 40c - 4d) \cup (A^* + 41c - 4d) \\
 I &:= (A^* + 47c - 5d) \cup (A^* + 48c - 5d) \\
 J &:= B \cup C \cup D \cup E \cup F \cup G \cup H \cup I
 \end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
 &\Delta_{(n+1)l-1}^{(0)}(x_1, \dots, x_m | z_1, \dots, z_{n+1} | z_{n+2}, \dots, z_N) \\
 = &\sum_{k=1}^{n+1} \frac{\prod_{i=n+2}^N (z_i - z_k q^{-2})}{\prod_{\substack{j=1 \\ j \neq k}}^{n+1} (z_k - z_j) q^{-1}} \Delta^{(nl)}(x_1, \dots, x_m | z_1, \dots, \overset{k}{\cdot}, z_{n+1} | z_k, z_{n+2}, \dots, z_N),
 \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
 \left| \|\varphi_{n,t}(b)\|^2 - \|b\|^2 \right| &= \left| \|\varphi_{n,t}(b)^* \varphi_{n,t}(b)\| - \|b^* b\| \right| \\
 &\leq \left| \|\varphi_{n,t-1}(b^*) \varphi_{n,t}(b)\| - \|\varphi_{n,e}(b^* b)\| \right| + \left| \|\varphi_{n,e}(b^* b)\| - \|b^* b\| \right| \\
 &\leq \left| \|\varphi_{n,t-1}(b^*) \varphi_{n,t}(b) - \varphi_{n,e}(b^* b)\| \right| + \left| \|\varphi_{n,e}(b^* b)\| - \|b^* b\| \right|
 \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
 [M_i, p_j] &= i\epsilon_{ijk} p_k, & [M_i, p_0] &= 0, \\
 [N_i, p_0] &= ip_i, & [p_\mu, p_\nu] &= 0, \\
 [N_i, p_j] &= i\delta_{ij} \left[\kappa c \sinh\left(\frac{p_0}{\kappa c}\right) e^{-\frac{p_0}{\kappa c}} + \frac{1}{2\kappa c} (\vec{p})^2 \right] - \frac{i}{\kappa c} p_i p_j.
 \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$X_1 := \{w \in W_{\text{per}}^{1,r}(\mathcal{S}) : w = 0 \text{ on } \{\zeta = -1\}\}$$

$$X_2 := L^\infty([-1, 0])$$

$$Y := \{u \in (W_{\text{per}}^{1,r'}(\mathcal{S}))^* : u = \mathcal{A} + \partial_\xi \mathcal{B}_1 + \partial_\zeta \mathcal{B}_2, \text{ for } \mathcal{A}, \mathcal{B}_1, \mathcal{B}_2 \in L^r(\mathcal{S})\},$$